Chapter 9: Carbon Compounds — Question Bank

1. Choose the Correct Option

1. Generally, the melting and boiling point of carbon compounds are found to be less than _____ $\text{^\circ C}$.
   - [ ] 300
   - [ ] 100
   - [ ] 200
   - [ ] 150
2. Number of valence electrons in a carbon atom is _____.
   - [ ] 4
   - [ ] 5
   - [ ] 1
   - [ ] 3
3. The bond between two oxygen atoms is _____ bond.
   - [ ] double
   - [ ] triple
   - [ ] single
   - [ ] none of these
4. The molecular masses of a carbon compound spread over a range of _____.
   - [ ] 10^{12}
   - [ ] 10^{14}
   - [ ] 10^{10}
   - [ ] 10^{13}
5. The unsaturated hydrocarbons containing a carbon-carbon double bond are called _____.
   - [ ] Alkenes
   - [ ] Alkanes
   - [ ] Alkynes
   - [ ] Alcohol
6. The phenomenon in which compounds having different structural formulae have the same molecular formula is called _____.
   - [ ] structural isomerism
   - [ ] catenation
   - [ ] homologous

- [ ] functional group

7. From the following hydrocarbon, _____ is the cyclic hydrocarbon.
   - [ ] isobutane
   - [ ] propyne
   - [ ] benzene
   - [ ] isobutylene
8. While going in an increasing order there is a rise in the molecular mass of the consecutive members of the homologous series by _____.
   - [ ] $14\ \mathrm{u}$
   - [ ] $15\ \mathrm{u}$
   - [ ] $16\ \mathrm{u}$
   - [ ] $17\ \mathrm{u}$
9. The general molecular formula for the homologous series of alkynes is _____.
   - [ ] $\mathrm{C}_n\mathrm{H}_{2n}$
   - [ ] $\mathrm{C}_n\mathrm{H}_{2n+2}$
   - [ ] $\mathrm{C}_n\mathrm{H}_{2n-2}$
   - [ ] $\mathrm{C}_n\mathrm{H}_{2n-1}$
10. _____ is one of the combustible components of L.P.G.
    - [ ] Methane
    - [ ] Ethane
    - [ ] Propane
    - [ ] Butanol
    - [ ] Ethene
11. At room temperature ethanol is _____.
    - [ ] solid
    - [ ] gas
    - [ ] plasma
    - [ ] liquid
12. Generally _____ is called spirit.
    - [ ] methanol
    - [ ] ethanol
    - [ ] propanol

- [ ] butanol

2. Find the Correlation

1. Solid : iodine :: _____ : bromine
2. $CH_3–CH_2–CHO$: propanal :: $CH_3–COOH$: _____
3. Ketone : $-CO-$ :: Ester : _____
4. Cyclohexane : Cyclic hydrocarbon :: Isobutylene : _____
5. Saturated hydrocarbon : Single bond :: Unsaturated hydrocarbon : _____
6. Saturated carbon compounds : blue flame :: Unsaturated carbon compounds : _____

3. Find Odd One Out

1. Find the odd one out and give its explanation:
   - Ethylene
   - Styrene
   - Propylene
   - Teflon
2. Find the odd one out and give its explanation:
   - Butane
   - Methane
   - Benzene
   - Ozone
3. Find the odd one out and give its explanation:
   - CH_4
   - C_2H_6
   - C_5H_{12}
   - $CaCO_3$
4. Find the odd one out and give its explanation:
   - C_2H_2
   - C_3H_8
   - C_2H_6

- CH_4

5. Find the odd one out and give its explanation:
   - C_2H_4
   - C_4H_{10}
   - C_3H_8
   - CH_4

4. Answer in One Sentence

1. Write the molecular formula of the given compound:
   - 1.1 Ethyl ethanoate
   - 1.2 Sodium ethanoate
   - 1.3 Sodium ethoxide
   - 1.4 Stearic acid
   - 1.5 Oleic acid
   - 1.6 Palmitic acid
2. Write the molecular formula of the given compound:
   - 2.01 Ethylene
   - 2.02 Benzene
   - 2.03 Acetic acid
   - 2.04 Propylene
   - 2.05 Acetylene
   - 2.06 Ethyl alcohol
   - 2.07 Acetone
   - 2.08 Propene
   - 2.09 Ethanol
   - 2.10 Ethanoic acid
   - 2.11 Isobutane
3. Draw electron dot structure and line structure for a given molecule:
   - 3.1 Hydrogen
   - 3.2 Oxygen
   - 3.3 Methane
   - 3.4 Nitrogen
   - 3.5 Ethene

5. Match the Columns

1. Match the pairs:

Group 'A'	Group 'B'
a. C_2H_6	1. Unsaturated hydrocarbon
b. C_2H_2	2. Molecular formula of an alcohol
c. CH_4O	3. Saturated hydrocarbon
d. C_3H_6	4. Triple bond

2. Match the columns:

Group A	Group B
1. Straight chain hydrocarbon	a) Benzene
2. Branched chain hydrocarbon	b) Propyne
3. Cyclic hydrocarbon	c) Isobutylene

3. Match the columns:

Group A	Group B
1) Ether	a) $-OH$
2) Ketone	b) $-O-$

3) Ester	c) $-CO-$
4) Alcohol	d) $-COO-$

6. Right or Wrong Sentence

1. The number of electrons in the valence shell of a carbon atom is 4.
2. Your body is made up of carbon.
3. Carbon compounds contain only open chains of carbon atoms.
4. In general, saturated compounds are more reactive than unsaturated compounds.
5. Benzene is a cyclic unsaturated hydrocarbon.
6. Cyclohexane is a branched chain type of hydrocarbon.
7. As one ascends in any homologous series, physical properties change gradually.
8. There are different general molecular formula for all members of the homologous series.
9. In LPG, butane is a flammable component.
10. Substances that can give oxygen to other substances are called reductant.
11. Potassium permanganate is an oxidizing compound in regular use.
12. Colorless ethanol is in liquid state at room temperature.
13. Ethanol is soluble in water in all proportions.
14. Ester is a sweet-smelling compound.
15. Two carbon atoms can always form one or two covalent bonds.

7. Write an Explanation

1. Write an Explanation: Alkane
2. Write an Explanation: Alkene
3. Write an Explanation: Alkyne

8. Write Scientific Reasons

1. Write scientific reason: Ethylene is an unsaturated hydrocarbon.
2. Write scientific reason: The flame appears yellow in the ignition of naphthalene.
3. Write scientific reason: The color of iodine disappears in the reaction between vegetable oil and tincture iodine.
4. Write scientific reason: Vegetable ghee is formed from the hydrogenation of vegetable oil in presence of nickel catalyst.
5. Write scientific reason: Carbon has the property of forming many compounds.
6. Write scientific reason: Benzene compounds are called aromatic compounds.

9. Solve the Following Questions

1. Explain the Catenation power.
2. Explain covalent bond with example.
3. Explain the Structural isomerism term with example.
4.
    - 4.1 What is an oxidising agent?
    - 4.2 Explain the following term with example: Reduction
5. Explain the concept of heteroatoms with the help of examples.
6.
    - 6.1 Explain the following reaction with example: Addition reaction
    - 6.2 Explain the following reaction with an example: Substitution reaction
    - 6.3 Explain the following reaction with an example: Esterification
    - 6.4 Explain the following reaction with an example: Saponification
7.
    - 7.1 What are catalysts?
    - 7.2 Give a chemical reaction in which a catalyst is used.
8. Write the characteristics of ethanol.
9. Write answer as directed: What is meant by vinegar and gasohol? What are their uses?
10. Write the uses of ethanol.
11. Write the characteristics of ethanoic acid.

10. Distinguish Between

1. Distinguish between: Saturated hydrocarbons - Unsaturated hydrocarbons
2. Distinguish between: Open chain hydrocarbons - Closed chain hydrocarbons
3. Distinguish between: Alkane - Alkene

11. Write Short Notes

1. Write a short note: Catenation power
2. Write a short note: Characteristics of carbon
3. Write a short note: Functional groups in carbon compounds
4. Write a short note: Homologous series
5. Write a short note: Aromatic hydrocarbons

12. Answer the Following

1. Complete the following table for the homologous series of alkanes:

Name	Molecular formula	Condensed structural formula	Number of carbon atom	Number of -CH2- units	Boiling point \text{^\circ C}
Methane	CH_4	CH_4	1	1	-162
Ethane	C_2H_6	$CH_3–CH_3$	2	2	-88.5
Propane	C_3H_8	$CH_3–CH_2–CH_3$	3	3	-42
Butane	C_4H_{10}	$CH_3–CH_2–CH_2–CH$	_____	_____	0
Pentane	C_5H_{12}	$CH_3–CH_2–CH_2–CH_2–CH$	_____	_____	36
Hexane	C_6H_{14}	$CH_3–CH_2–CH_2–CH_2–CH_2–CH$	_____	_____	69

2. Complete the following table for homologous series of alcohols:

Name	Molecular formula	Condensed structural formula	Number of carbon atom	Number of -CH2- units	Boiling point \text{^\circ C}
Methanol	CH_4O	$CH_3 –$	1	1	63
Ethanol	C_2H_6O	$CH_3–CH_2 –$	2	2	78
Propanol	C_3H_8O	$CH_3–CH_2–CH_2 –$	_____	_____	97
Butanol	$C_4H_{10}O$	$CH_3–CH_2–CH_2–CH_2–OH$	_____	_____	118

3. Complete the following table for homologous series of Alkenes:

Name	Molecular formula	Condensed structural formula	Number of carbon atom	Number of -CH2- units	Boiling point \text{^\circ C}
Ethene	C_2H_4	$CH_2=$	2	0	-102
Propene	C_3H_6	$CH_3–CH=$	3	1	-48
1-Butene	C_4H_8	$CH_3–CH_2–CH=$	_____	_____	-6.5
1-Pentene	C_5H_{10}	$CH_3–CH_2–CH_2–CH=$	_____	_____	30

4. Complete the given chart by writing the correct functional group of carbon compounds:
(Options: Ester, Aldehyde, Ketone, Carboxylic acid, Alcohol, Ether)

Structural Formula Unit	Functional Group
$-O-$	
$-C(=O)-$	
$-C(=$	
$-C(=O)-O-$	
$-O-$	
$-C(=O)-$	
$-N(-H)-$	

5. Complete the following table by writing the correct structural formula and molecular formula:

Straight chain of carbon atoms	Structural formula	Molecular formula	Name
$C-$	_____	_____	Ethane
$C-C-C-$	_____	_____	Butane
$C-C-C-C-$	_____	C_7H_{16}	_____
$C-C-C-C-$	_____	C_8H_{18}	_____

6. Complete the following table by writing the IUPAC name of the carbon compound:

Sr. No.	Common name	Structural formula	IUPAC Name
1.	ethylene	$CH_2=$	
2.	acetylene	$HC\equiv$	
3.	acetic acid	CH_3-	
4.	methyl alcohol	CH_3-	
5.	ethyl alcohol	CH_3-CH_2-	
6.	acetaldehyde	CH_3-	
7.	acetone	CH_3-CO-	
8.	ethyl methyl ketone	$CH_3-CO-CH_2-$	

9.	ethyl amine	$CH_3 - CH_2 -$	
10.	n-propyl chloride	$CH_3 - CH_2 - CH_2 -$	

7. Complete the following activity:
   - Boiling point of ethanol $\rightarrow$
   - General name of ethanol $\rightarrow$
   - Use of ethanol $\rightarrow$
   - Boiling point of ethanoic acid $\rightarrow$
   - Melting point of pure ethanoic acid $\rightarrow$
8. Complete the following activity.
9. Observe the structural formula and answer the following questions:
   1. Write the name of the given hydrocarbon.
   2. The given hydrocarbon is included in which type of hydrocarbon?
   3. What is the kind of compounds with the above characteristic structure called?
10. Complete the following chart by using examples given in brackets:
    (isobutylene, cyclohexane, propane, cyclohexene, cyclopentane, benzene, propyne, isobutane, propene)

Straight chain hydrocarbons	Branched chain hydrocarbons	Cyclic hydrocarbons

11. Identify and write the names of the hydrocarbons for the given structural formula *(using options from brackets: isobutylene, cyclohexane, propene, cyclohexene, cyclopentane, benzene, propyne, isobutane, propane)*:
    - 11.1 $H_3C - CH = CH_2$
    - 11.2 $H_3C - CH_2 - CH_3$
    - 11.3 $HC \equiv C - CH_3$
    - 11.4 $CH_3 - CH(CH_3) - CH_3$
    - 11.5 Identify structural formula 11.5 from options.
    - 11.6 Identify structural formula 11.6 from options.

- 11.7 Identify structural formula 11.7 from options.
- 11.8 Identify structural formula 11.8 from options.
- 11.9 Identify structural formula 11.9 from options.

13. Answer the Following Questions

1. Observe the figure and write the answers to the following questions:
    1. Write the name of the reaction shown in the figure.
    2. Write the above chemical reaction in the form of a balanced equation.
    3. Write the name of the product produced in the above reaction, write a use.
    4. Write the name of the catalyst used in the above reaction.
2. Write the answers to the questions by observing the figure:
    1. Write the chemical reaction shown in the figure in the form of a balanced equation.
    2. Write the name of the gas coming out of the large test tube in the chemical reaction.
    3. Why do small bubbles appear in the small test tube?
    4. What is the change in colour of lime water?